

MIRAbot: A Rearview Mirror Driving Assistant for Semi-Autonomous Vehicles

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Figure 1: MIRAbot prototype installed in a car

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Abstract

This video introduces MIRAbot, a reimagined rearview mirror designed as a driving assistant for SAE Level 3 semi-autonomous

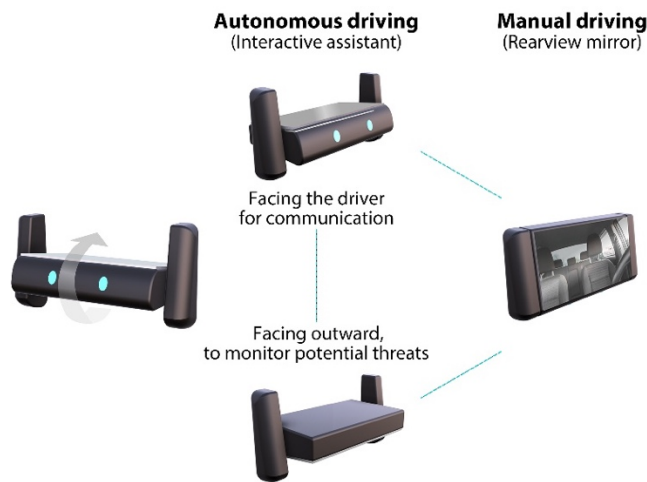


Figure 2: MIRAbot rotates to support three modes: (top middle) facing the passenger for communication, (right) functioning as a rearview mirror, and (bottom middle) facing outward to scan for potential threats.

vehicles (see Figure 1). While Level 3 vehicles allow hands-free, eyes-off driving in specific conditions, drivers must remain ready to take control when necessary [3]. Despite their convenience, these vehicles face challenges in earning drivers' trust and achieving widespread adoption. MIRAbot seeks to enhance the semi-autonomous driving experience by making it more trustworthy and accessible to a diverse range of users. MIRAbot seamlessly switches between functioning as a standard rearview mirror and an interactive assistant, supporting manual driving, autonomous driving, and the transitions between them (see Figure 2). Building on prior research into robotic rearview mirror ornaments for handover pre-alerts [1, 2], MIRAbot enriches the in-car experience with expressive eyes, anthropomorphic movements, and voice prompts. This video highlights MIRAbot's design, interaction features, and functionality, showcasing its potential to make semi-autonomous driving more supportive, enjoyable, and accessible for all.

CCS Concepts

• **Human-centered computing**; • **Interaction design**; • **Interaction design process and methods**; • **Interface design prototyping**;

Keywords

Semi-autonomous vehicle, Society of Autonomous Engineers (SAE) Level 3, Robotic object, Vehicle robot, Conversational voice agent, In-vehicle intelligent agent (IVIA), Handover, Takeover Request (TOR)

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